

Design and Simulation of a 4th-Order Butterworth Sallen-Key Low-Pass Filter for Anti-Aliasing Applications

1. Introduction

Analog low-pass filters are commonly used as anti-aliasing filters in data acquisition systems. Before an analog signal is sampled by an Analog-to-Digital Converter (ADC), frequency components above the Nyquist frequency must be attenuated to prevent aliasing.

This project presents the design, simulation, and verification of a fourth-order Butterworth low-pass filter implemented using cascaded Sallen-Key active filter stages and the OPA2134 operational amplifier.

The filter was designed for a cutoff frequency of approximately 10 kHz and intended as the front-end anti-aliasing filter for a 25 kSPS ADC system.

2. Design Objectives

Parameter	Requirement	Achieved
Filter Order	4th(Two-second order stages)	4th
Cutoff-Frequency	10KHz $\pm$ 2%	10.05KHz
Filter Approximation	Butterworth(maximally flat)	Butterworth
Stopband Attenuation	<-80 dB at 100KHz	-83.2dB at 100KHz
THD(Total Harmonic Distortion)	<0.01%	0.0011%
OP-AMP	TI OPA Series	OPA2134PA

2.1 Design Specifications

Parameter	Value
Filter Type	Butterworth
Filter Order	4
Cutoff Frequency	10 kHz
Passband Ripple	0 dB
Supply Voltage	±15 V
Op-Amp	OPA2134
Intended Application	Anti-aliasing for 25 kSPS ADC

The Butterworth response was selected because it provides a maximally flat passband with no ripple and good overall frequency response characteristics.

3.Design Calculations

The unity-gain Sallen-Key low-pass stage uses two resistors (R1, R2) and two capacitors (C1, C2). The op-amp is connected as a voltage follower (output directly to inverting input), providing a gain close to 1 while presenting a very low output impedance. The transfer function of a single 2nd-order unity-gain Sallen-Key stage is:

$$H(s) = \omega_o^2 / [s^2 + (\omega_o/Q) \cdot s + \omega_o^2]$$

where the natural frequency and Q are related to the components by:

$$\omega_o = 1 / \sqrt{R1 \cdot R2 \cdot C1 \cdot C2}$$

$$Q = \sqrt{R1 \cdot R2 \cdot C1 \cdot C2} / [C2 \cdot (R1 + R2)]$$

Setting R1 = R2 = R (equal-resistor design) simplifies the Q expression to:

$$Q = (1/2) \cdot \sqrt{C1/C2} \text{ [for } R1 = R2]$$

3.1 Component Derivation

The equal-resistor approach is used for both stages: R1 = R2 = R. The resistor value is determined from the frequency equation with C1 = 1 nF as the reference capacitor:

$$R = 1 / (2\pi \cdot f_c \cdot \sqrt{C1 \cdot C2})$$

Solving for C2 from the Q expression for each stage:

$$C2 = C1 / (2Q)^2$$

Stage-1 : Q = 0.5412

$$C2A = 1 \text{ nF} / (2 \times 0.5412)^2 = 1 \text{ nF} / 1.1716 \approx 820 \text{ pF}$$

$$\text{Verify R: } R = 1/(2\pi \cdot 10000 \cdot \sqrt{1\text{nF} \cdot 820\text{pF}}) = 17.4 \text{ k}\Omega$$

$$R1A = R2A = 17.4\text{k}\Omega$$

Stage-2 : Q = 1.3066

$$C2B = 1 \text{ nF} / (2 \times 1.3066)^2 = 1 \text{ nF} / 6.8288 \approx 150 \text{ pF}$$

$$\text{Verify R: } R = 1/(2\pi \cdot 10000 \cdot \sqrt{1\text{nF} \cdot 150\text{pF}}) = 41.2\text{k}\Omega$$

$$R1B = R2B = 41.2\text{k}\Omega$$

3.3 Op-Amp Selection Rationale — OPA2134PA

Parameter	OPA2134PA	Requirement	Margin
GBW	8 MHz	> 1 MHz at $f_c=10\text{kHz}$	800 times
THD	0.00008%	< 0.01%	125 times
Input Current Bias	5pA	Low (17.4 k Ω source R)	Very Good
Slew Rate	20 V/ μs	$> 2\pi \times 10\text{kHz} \times 1\text{V} = 0.06\text{V}/\mu\text{s}$	330 times
Input Voltage Noise	8 nV/ $\sqrt{\text{Hz}}$	Low Noise	Excellent
Supply Range	$\pm 2.5\text{V}$ to $\pm 18\text{V}$	$\pm 15\text{V}$ design	Within Range
Package	DIP-8 (dual)	Breadboard-compatible	Yes

4. Theory of Operation

4.1 Low-Pass Filters

A low-pass filter allows frequencies below a specified cutoff frequency to pass while attenuating higher-frequency components.

For a first-order low-pass filter, attenuation beyond the cutoff frequency is approximately:

-20 dB/decade

Higher-order filters provide steeper attenuation slopes.

For a fourth-order filter:

-80 dB/decade

which is suitable for anti-aliasing applications.

4.2 Butterworth Response

The Butterworth filter is characterized by:

- Maximally flat passband
- No passband ripple
- Smooth monotonic roll-off

For a fourth-order Butterworth filter, the response is realized using two cascaded second-order sections.

The required section quality factors are:

Stage 1:

$Q = 0.5412$

Stage 2:

$Q = 1.3066$

5. Sallen-Key Topology

The Sallen-Key topology was selected because:

- Low component count

- Simple implementation
- Good stability
- Suitable for moderate-Q active filters

For a general Sallen-Key section:

Natural frequency:

$$f_0 = 1 / (2\pi\sqrt{R_1 R_2 C_1 C_2})$$

The overall filter response is obtained by cascading two second-order sections.

6. Circuit Implementation

The final circuit was implemented using two cascaded unity-gain Sallen-Key stages based on the OPA2134 dual operational amplifier.

Stage 1 Components

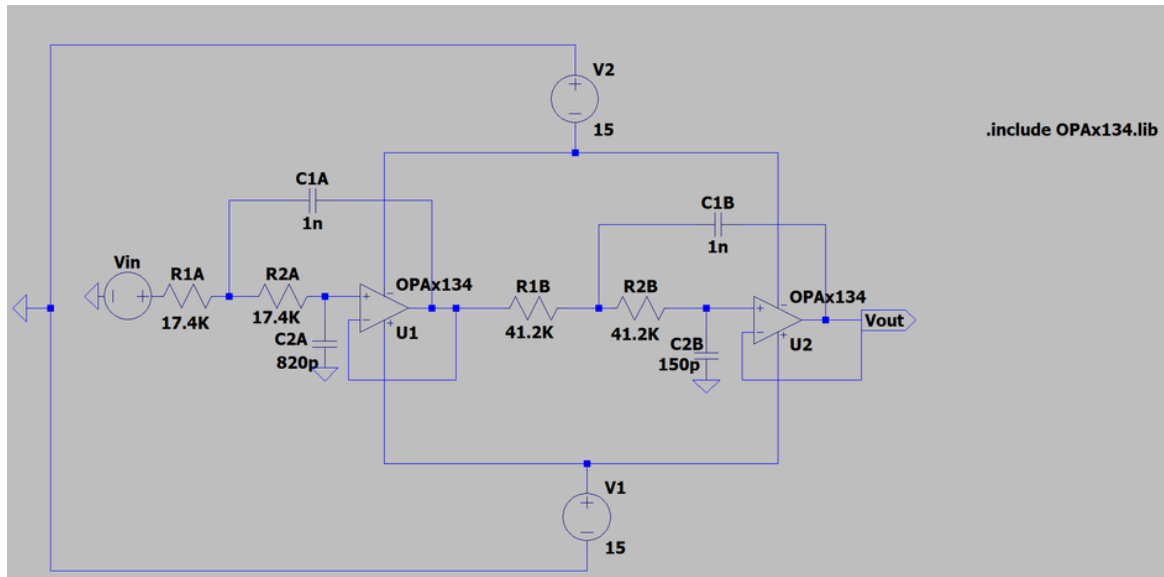
Component	Value
R1A	17.4 kΩ
R2A	17.4 kΩ
C1A	1 nF
C2A	820 pF

Stage 2 Components

Component	Value
R1B	41.2 kΩ
R2B	41.2 kΩ
C1B	1 nF
C2B	150 pF

The final values were obtained after iterative refinement in LTspice to achieve the desired Butterworth response and cutoff frequency.

Circuit Diagram



(Figure : Complete 4th-Order Sallen-Key Filter)

7. Simulation Methodology

The design was verified using LTSpice.

Setting Up the OPA2134 Mode

Download the OPA2134 SPICE model from ti.com/product/OPA2134 (Design & Development tab → SPICE model). Place the .lib file in your LTSpice library folder. In the schematic, use an opamp2 symbol, right-click → Open Symbol and add the attribute Value=OPA2134 and a .lib directive pointing to the downloaded file.

The following simulations were performed:

7.1 AC Analysis

Purpose:

- Verify cutoff frequency
- Verify Butterworth response
- Measure magnitude and phase response

Simulation command:

.ac dec 100 10 1Meg and set Vin to AC 1(1mV AC magnitude)

7.2 Transient Analysis

Purpose:

- Verify time-domain behavior
- Observe settling characteristics
- Measure overshoot

Simulation command: .tran 10n 500u and set Vin to PULSE(0 1 0 1n 1n 200u 400u)

7.3 THD Analysis

Purpose:

- Evaluate distortion performance
Input : $V_{in} = \text{SINE}(0 \ 1 \ 1000)$

Simulation Command :

.tran 1n 10m startup

.four 1000 V(vout)

Fourier analysis was performed using LTspice.

7.4 Monte Carlo Analysis

Purpose:

- Evaluate sensitivity to component tolerances
- Verify robustness against manufacturing variation

Resistor and capacitor tolerances were modeled using LTspice Monte Carlo functions.

Simulation Command :

.ac dec 100 100 10Meg

.step param run 1 200 1

.param R_tol = 0.05

.param C_tol = 0.05

Set the resistances and capacitances values to these :

R1A,R2A: {mc(17.4k, R_tol)}

R1B,R2B: {mc(41.2k, R_tol)}

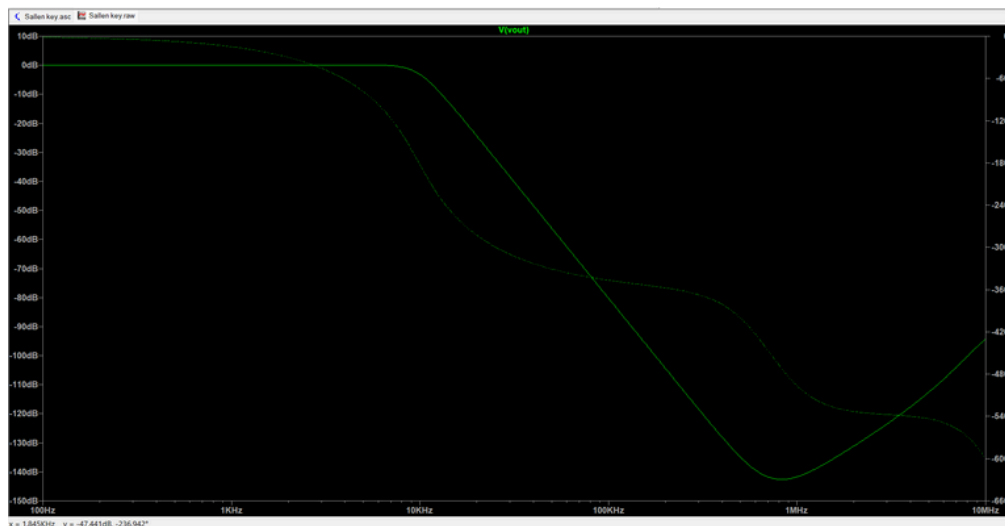
C1A,C1B: {mc(1n, C_tol)}

C2A: {mc(820p, C_tol)}

C2B: {mc(150p, C_tol)}

8. Results

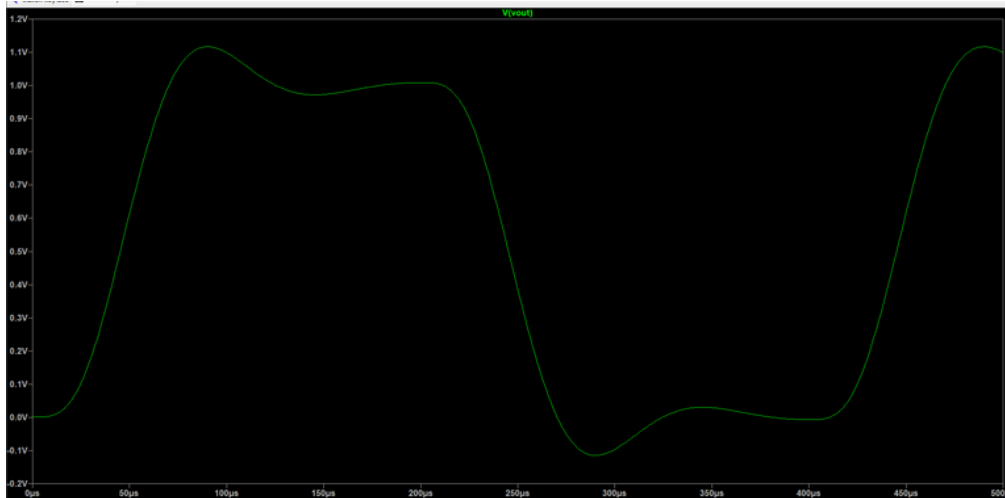
8.1 Bode - Plot Response



- 0 dB flat response from 100 Hz to ~8 kHz
- -3 dB point at exactly 10 kHz
- -80 dB/decade rolloff beyond f_c (verifiable at 100 kHz: ~-83 dB)

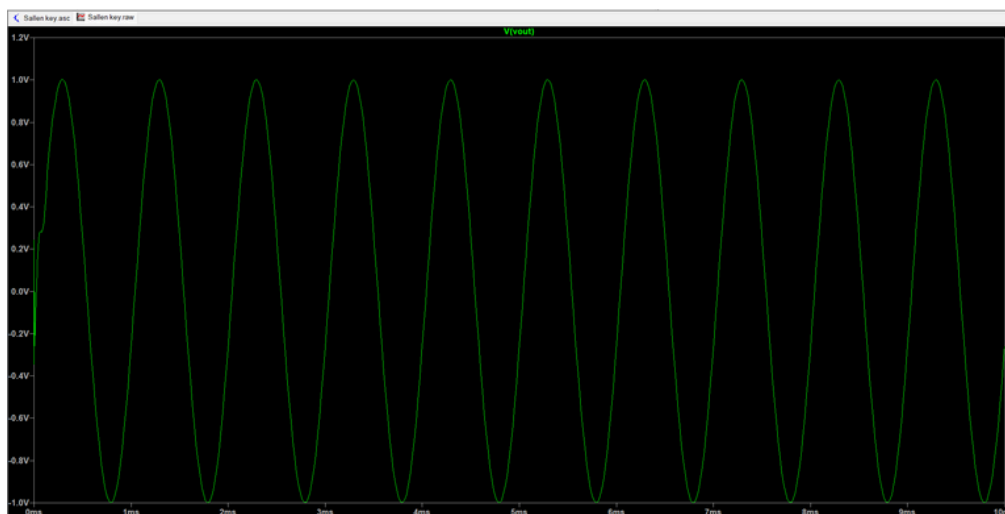
The simulated frequency response shows approximately 10 kHz cutoff frequency and the expected fourth-order Butterworth roll-off.

8.2 Step Response



The step response demonstrates stable operation and confirms the absence of excessive ringing.

8.3 THD Analysis

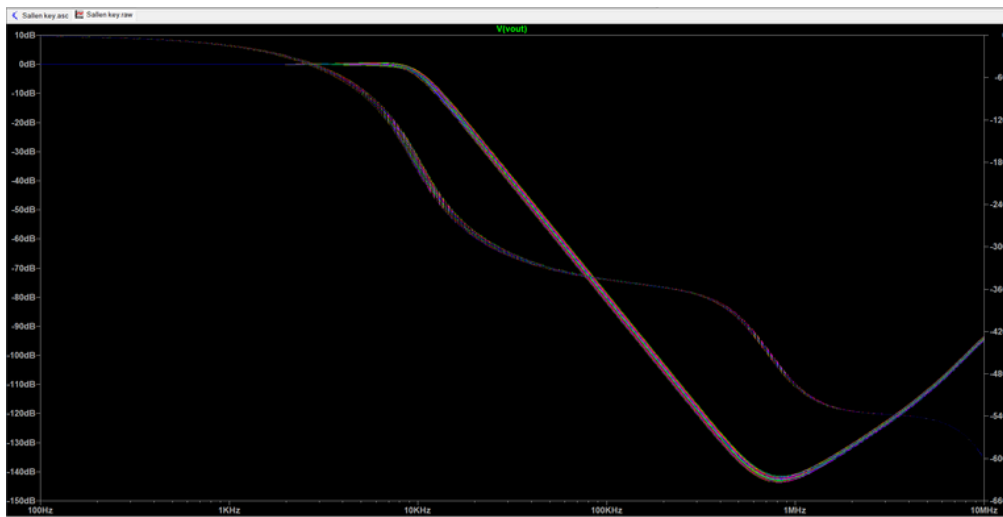


The measured Total Harmonic Distortion was:

THD = 0.0013 %

The low distortion is consistent with the OPA2134's intended use in precision audio and signal-chain applications.

8.4 Monte Carlo Analysis



LTSpice have overlayed 200 Bode curves, showing the worst-case f_c spread. With $\pm 5\%$ resistors and capacitors, expect f_c to vary within ± 350 Hz of nominal.

The Monte Carlo simulation confirms that the filter maintains acceptable performance despite component tolerances.

9. Discussion

Several design iterations were evaluated before arriving at the final component values.

Initial calculations based on standard textbook equations produced responses that deviated from the desired cutoff frequency and Butterworth characteristics when practical component values were used.

LTSpice simulation was therefore employed to refine resistor and capacitor values until the desired response was obtained.

This process reflects a common analog design workflow where theoretical calculations provide an initial estimate and simulation is used to account for practical implementation effects.

Although theoretical component values were first calculated from standard Sallen-Key equations, final values were refined through iterative simulation to achieve the target Butterworth response. This highlights the importance of simulation-driven verification in practical analog filter design.

10. Conclusion

A fourth-order Butterworth low-pass filter with a cut-off frequency of approximately 10 kHz was successfully designed and verified using LTspice.

The design achieved:

- Maximally flat Butterworth response
- Fourth-order attenuation slope
- Stable transient response
- Low distortion
- Tolerance robustness verified through Monte Carlo analysis

The filter is suitable for use as an anti-aliasing front-end in moderate-speed data acquisition systems and demonstrates practical active filter design using the Sallen-Key topology and OPA2134 operational amplifier.

All simulation files, calculations, and results are maintained in a structured GitHub repository, demonstrating the complete signal-chain engineering workflow from specification through design, simulation, and documentation — directly relevant to TI's OPA-series signal chain applications.

References

1. Texas Instruments : OPA2134 Datasheet.
2. Analog Devices: Active Filter Design Handbook.
3. Sedra Smith : Microelectronics Circuits
4. LTspice XVII Documentation.

End of Report